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# Birdwatching

**Description:**
*Birdwatching* is a VR exploration game where you play as a field biologist studying birds in a post-nuclear area. In VR with motion controls, you use your left hand to hold a camera and take pictures, and your right hand to feed birds using a feeding stick.

The main goal of the game is to find different birds, take photos of them, and fill out a bingo book. When you take a picture, it automatically gets added to the book. Players are encouraged to move carefully and interact gently so they don't scare the birds away.

The game focuses on relaxing exploration and observing wildlife. A stretch goal is to add a co-op mode where players can explore and collect birds together.

**Story:**
The player is a nuclear recovery biologist (ornithologist if we stick with birds specifically) on an expedition to the site of a nuclear detonation with the goal of documenting the remaining bird/wildlife population and study any changes.

**Art Style:**

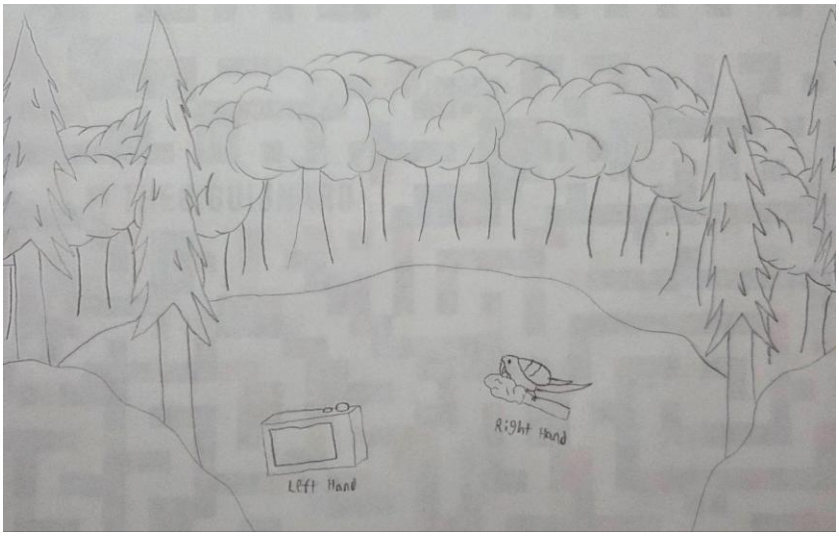
- PS1 Style graphics.

- Similar to Metal Gear, Resident Evil, Silent Hill, Tomb Raider, etc.
- Nostalgic and simplistic. Fits with the hobby of birdwatching.

- Much easier style to work with for the given timeframe.

**Mechanics:**

- The left hand holds a camera, the right holds a feeding stick.




- Hover controller over birds for points and money
- Rarer birds are higher in points and money
- Use money to buy equipment that lures rarer birds
- Player can't lose
- Buy feeding sticks so birds can land on your hand.

- When bird lands - be careful, be gentle with your motion controls not to scare it away

- Crows thank you for the food with coins
- Bird textures variations for more bird variety
- Co op - stretch goal
- Bingo book - 3D, snapshot system where camera photos get immediately reflected into the bird bingo squares
- Gold stickers for bingo scores

**Audience:**

The intended audience for this game is teens and adults. The hobby is very popular amongst that demographic so by combining the two topics we allow for a familiar yet unique experience. Furthermore both topics are already similar where both involve looking out for various birds. This VR game differs from regular birdwatching by having the player get up close with the birds and interacting with them through events like feeding.

**Realistic Scope Assessment**

Given the semester timeframe, we have to keep it realistic and properly scope the project down to a single, highly polished woodland environment. Instead of a massive open world, the focus will be a simple controlled scene, the bingo book, and perfecting the physical camera interactions and the bird-feeding mechanics. We plan to implement at least 1 extra bird texture for each bird species to ensure the core gameplay loop (finding, photographing, and logging) functions well and with some variety. Hopefully, without bugs before the May 7th deadline. Co-op multiplayer will remain strictly a stretch goal.

**Independent Research Topic:** For the required mechanic not covered in class, we will research how to use Unity's "Render Texture" feature to create a functional in-game digital camera. We need to learn how to script the camera so that when the player pulls the VR trigger, it captures a 2D snapshot of the 3D world and immediately applies that image as a texture onto the 3D Bingo Book pages. This will be a big challenge, but it will be critical for our game's feedback loop.

**Assets we plan to use:**

https://assetstore.unity.com/packages/3d/vegetation/the-illustrated-nature-sample-161188
Provides stylized, high-quality trees and grass to create the immersive outdoor environment the player will explore.

https://assetstore.unity.com/packages/3d/characters/animals/birds/living-birds-15649
Adds animated, realistic birds to act as the primary wildlife the player will track and interact with.

https://assetstore.unity.com/packages/3d/props/digital-camera-227755
Serves as the player's main interactive tool, allowing them to physically frame and take photos using their VR headset and controllers.

https://assetstore.unity.com/packages/3d/props/low-poly-bird-nests-229812
Provides hidden environmental details or specific exploration objectives to place throughout the forest canopy.

https://assetstore.unity.com/packages/3d/environments/low-poly-environment-nature-free-lowpoly-medieval-fantasy-series-187052
Supplies essential building blocks like rocks and terrain pieces to quickly block out the foundation of the level design map.

https://assetstore.unity.com/packages/3d/props/books-essentials-362582
Will be scripted into an interactive, physical bingo book to journal the bird photos and achieve stars.

https://assetstore.unity.com/packages/3d/environments/environment-forest-low-poly-3d-295608
Adds necessary variety to the woodland areas, creating dense, diverse habitats for the birds to hide in. This might end up being the player’s spawn point, working as a gradual introduction into the forest.

https://assetstore.unity.com/packages/vfx/shaders/low-poly-wind-182586
Adds a wind effect to make the static foliage sway, providing the continuous visual feedback needed to keep a stereoscopic world feeling alive.